

WMDR2-core

Package in package 'Model'

WMDR2-core

Version 0.2.4 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-26

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

wmdr2-core v0.2.x

v0.2.0

- time (a temporal extent) shall be realized as time:interval (unchanged from v.0.1.13)
- Following the OGC Features model, geometry = geolocation = coordinates + elevation
- All temporal<Name> elements are renamed to historical<Name>.
- historical<Name> elements other than historicalLocation shall be realized as arrays of dictionary objects for (begin) date, and the values of <Name>, e.g.,

```
historicalTerritory = [{
  "date": "2000-01-01",
  "territory": "<ISO1>"
},
{
  "date": "2024-07-13",
  "territory": "<ISO2>"
}]
```

- "observation" describes time-series (collections) of (individual) observations / measurements.
- "observation"s are acquired using a "historicalObservingSchedule", which has a (begin) date and refers to a schedule / coverage.
- "deployment" is removed as a concept, most attributes are directly under "observation". - "instrument" links directly to "observation". In alignment with OMS/STA, there is a link to "facility" via "deplyoment", but this is not expected to be used at this point.
- "observation"s can be grouped to an "observationSets" (NB: This is a test and not guaranteed to exist in future.)

v0.2.1

- "observation" has been renamed to "observations", because it describes collections (=time-series) of (individual) observations / measurements.
- "observations" are acquired using a "historicalObservingSchedule", which has a (begin) date and refers to a schedule / coverage.
- The concept of deployment has been developed into "historicalDeployment", which is now a more focussed and intuitive concept. It describes the placement at a specific date of an instrument at a location (=geometry) that can generate observations. It describes the "exposure" of the instrument as well as its "operatingStatus" and "serialNumber" from that date onwards.
- "observations" can be grouped to an "observationSets" (NB: This is experimental and not guaranteed to exist in future.)
- "reporting" and "historicalReporting" are modeled to allow reuse of a reporting template. This is similar to the "schedule" / "historicalObservingSchedule" concept.

v0.2.2

- clean up attribute names, consistent singular naming within historical<Name> classes,

v0.2.3

- Class names capitalized, except if class is re-usable.
- prefix "historical" removed as not needed (TT-WMDR2, 2026-06-24), the "date" element makes this clear
- "observations" renamed to "ObservationSeries" to avoid mis-interpretation.
- update cardinalities on instrument.manufacturer to [0..1], instrument.model to [0..1]
- remove ObservationSeries.time (can be inferred from Deployments.)
- rename ObservingSchedule to ObservingProcedure and add "strategy", rename schedule to observingSchedule, rename sampling to samplingFrequency, aggregation to aggregationInterval
- rename Reporting to ReportingProcedure and add strategy; remove aggregation (a misnomer and different concept, used as an alias of temporalReportingInterval), add ReportingProcedure.reportingSchedule to allow regular but

v0.2.4

- correct facility.Deployments cardinality from 0..* to 1..*
- correct ObservingSeries.Deployments cardinality from 0..1 to 1..*
- move ObservingSeries.ReferenceSurface and .verticalDistance... to Deployment
- move instrument.observingMethod to ObservingMethod.observingMethod and correct cardinalities
- remove ObservationSeries.time as redundant
- officialStatus is a property of ObservationSeries and not time-bound
- move ObservationSeries.sourceOfObservation to Deployment.sourceOfObservation
- remove experimental (blue) classes

Version 0.2.1

ijkl created on 2026-03-06. Last modified 2026-06-26

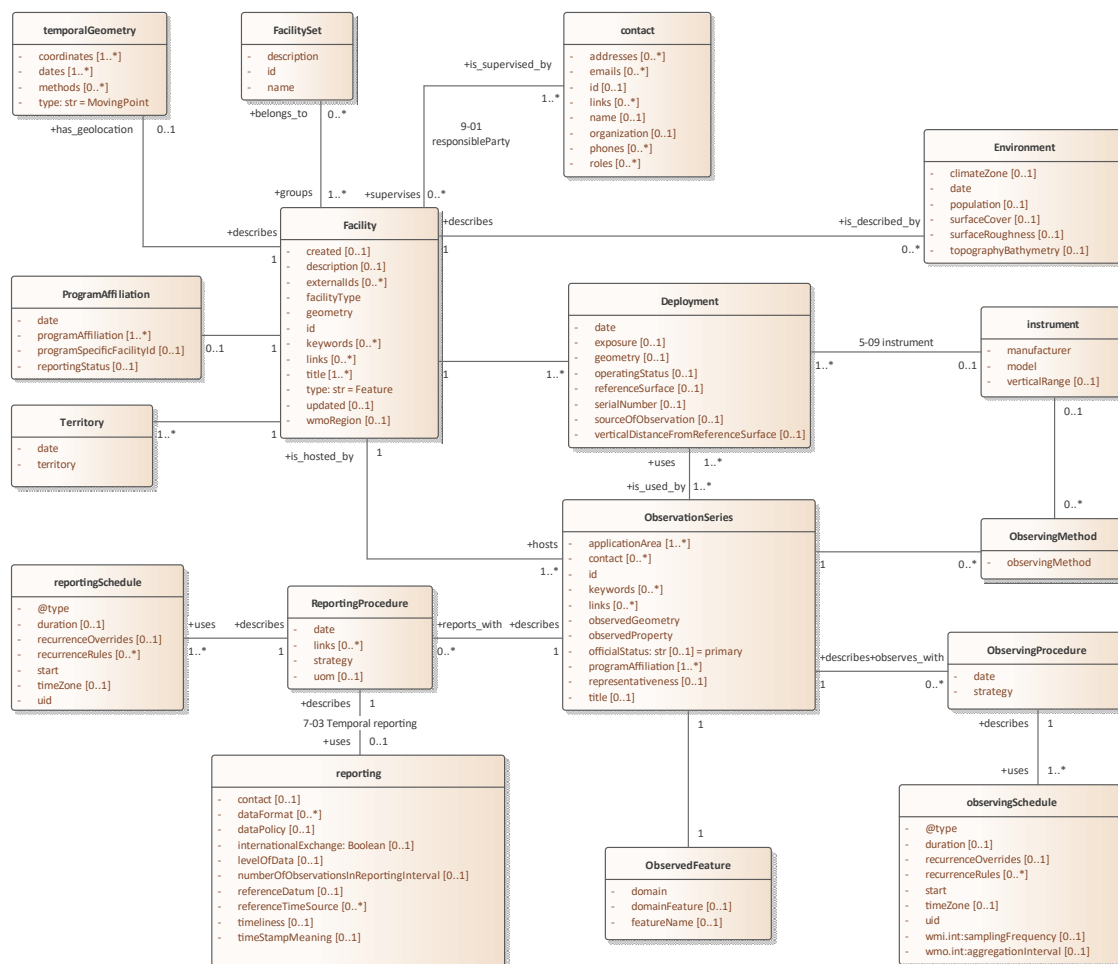


Figure 1: WMDR2-core

contact

Class in package 'WMDR2-core'

WMDR2 core JSON class: contact.

contact
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-25

ATTRIBUTES	
◆ addresses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Postal addresses. [Is static True. Containment is Not Specified.]	
◆ emails : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Email contacts. [Is static True. Containment is Not Specified.]	
◆ id : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional stable contact identifier. [Is static True. Containment is Not Specified.]	
◆ links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Online resources associated with the contact. [Is static True. Containment is Not Specified.]	
◆ name : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Either a full name, or a sequence of name parts. Examples: Dr. Jörg Klausen (this corresponds to FN in vcard) Klausen;Jörg;Dr.; (this corresponds to N in vcard) [Is static True. Containment is Not Specified.]	
◆ organization : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Example: ORG:Federal Office of Meteorology and Climatology MeteoSwiss [Is static True. Containment is Not Specified.]	
◆ phones : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Phone contacts. [Is static True. Containment is Not Specified.]	
◆ roles : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES

Specific contact roles such as owner, pointOfContact, principalInvestigator.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) Facility
Cardinality: [0..*]

Target: Public is_supervised_by (Class) contact
Cardinality: [1..*]

Deployment

Class in package 'WMDR2-core'

Deployment

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-22. Last modified 2026-06-25

ATTRIBUTES

 date : Private

[Is static True. Containment is Not Specified.]

 exposure : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Values from <http://codes.wmo.int/wmdr/Exposure>

[Is static True. Containment is Not Specified.]

 geometry : Private

Alias: temporalGeometry

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

geometry is the OGC Records name for geolocation.

Meaningful only for fixed facilities, do not use for mobile facilities.

[Is static True. Containment is Not Specified.]

 operatingStatus : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

values refer to <http://codes.wmo.int/wmdr/InstrumentOperatingStatus>

[Is static True. Containment is Not Specified.]

 referenceSurface : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Linked to heightAboveReferenceSurface, from <http://codes.wmo.int/wmdr/ReferenceSurfaceType>

[Is static True. Containment is Not Specified.]

ATTRIBUTES

serialNumber : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
 [Is static True. Containment is Not Specified.]

sourceOfObservation : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
 Value from <http://codes.wmo.int/wmdr/SourceOfObservation>.
 If sourceOfObservation=="humanObservation", several attributes of "observations" are expected to be null.
 [Is static True. Containment is Not Specified.]

verticalDistanceFromReferenceSurface : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
 implement as an array of objects:
 {"value": float, "uom": str}, whereby "uom" from <http://codes.wmo.int/wmdr/unit>
 [Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 5-09 instrument
 Source: Public (Class) Deployment
 Cardinality: [1..*]
 Target: Public (Class) instrument
 Cardinality: [0..1]

Association (direction: Unspecified)
 Source: Public (Class) Facility
 Cardinality: [1]
 Target: Public (Class) Deployment
 Cardinality: [1..*]

Association (direction: Unspecified)
 Source: Public is_used_by (Class) ObservationSeries
 Cardinality: [1..*]
 Target: Public uses (Class) Deployment
 Cardinality: [1..*]

Environment

Class in package 'WMDR2-core'

Facility-environment object containing climate, surface-cover, population, and topography/bathymetry histories.

Environment
 Version 1.0 Phase 1.0 Proposed
 ChatGPT created on 2026-06-02. Last modified 2026-06-25

ATTRIBUTES

climateZone : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES	
	[Is static True. Containment is Not Specified.]
 date : Private	[Is static True. Containment is Not Specified.]
 population : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Population in 10 and 50 km radius around the facility	[Is static True. Containment is Not Specified.]
 surfaceCover : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 surfaceRoughness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 topographyBathymetry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) Facility Cardinality: [1]	Target: Public is_described_by (Class) Environment Cardinality: [0..*]

Extension

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

Extension
 Version 1.0 Phase 1.0 Proposed
 jkl created on 2026-03-11. Last modified 2026-06-26

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservationSeries Cardinality: [1..*]	Target: Public uses (Class) Extension Cardinality: [0..1] ObservingProcedure(s)

Facility

Class in package 'WMDR2-core'

Facility properties payload of the WMDR2 full record. Root Features geometry/time are modeled on fullRecord, not here.

Facility
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES	
 created : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record creation date/time metadata.	[Is static True. Containment is Not Specified.]
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Facility description.	[Is static True. Containment is Not Specified.]
 externalIds : Private Alias: 3-06 (primary & additional) WSIs Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) External identifiers such as WMO:WIGOS.	[Is static True. Containment is Not Specified.]
 facilityType : Private Alias: 3-04 facilityType Compact WMDR facility type code value.	[Is static True. Containment is Not Specified.]
 geometry : Private Alias: 3-07 current geolocation geometry = geolocation = coordinates + elevation	[Is static True. Containment is Not Specified.]
 id : Private Alias: 3-06 (primary) WSI to prefixed with name space "facility:"	[Is static True. Containment is Not Specified.]
 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES

Discovery/search keywords retained in the core JSON.	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links.	[Is static True. Containment is Not Specified.]
 title : Private Alias: 3-03 facilityName Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) properties.title = 3-03 facilityName	[Is static True. Containment is Not Specified.]
 type : str Private = Feature Facility properties type; const facility.	[Is static True. Containment is Not Specified.]
 updated : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record update date/time metadata.	[Is static True. Containment is Not Specified.]
 wmoRegion : Private Alias: 3-01 wmoRegion Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified) Source: Public describes (Class) Facility Cardinality: [1]	Target: Public is_described_by (Class) Environment Cardinality: [0..*]
 Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) Territory Cardinality: [1..*]
 Association (direction: Unspecified) Source: Public groups (Class) Facility Cardinality: [1..*]	Target: Public belongs_to (Class) FacilitySet Cardinality: [0..*]

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public describes (Class) Facility Cardinality: [1]	Target: Public has_geolocation (Class) temporalGeometry Cardinality: [0..1]
 Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) ProgramAffiliation Cardinality: [0..1]
 Association (direction: Unspecified) Source: Public is_hosted_by (Class) Facility Cardinality: [1]	Target: Public hosts (Class) ObservationSeries Cardinality: [1..*]
 Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) Deployment Cardinality: [1..*]
 Association (direction: Unspecified) 9-01 responsibleParty Source: Public supervises (Class) Facility Cardinality: [0..*]	Target: Public is_supervised_by (Class) contact Cardinality: [1..*]

FacilitySet

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

FacilitySet
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES	
 description : Private [Is static True. Containment is Not Specified.]	
 id : Private [Is static True. Containment is Not Specified.]	
 name : Private Alias: title [Is static True. Containment is Not Specified.]	

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public groups (Class) Facility
Cardinality: [1..*]

Target: Public belongs_to (Class) FacilitySet
Cardinality: [0..*]

instrument

Class in package 'WMDR2-core'

Describes instrument capabilities, not a specific instrument instance.

For semantic consistency, terms like observedProperty and observedGeometry are used, even though they just describe the instrument capabilities, not the actual observations. With this, instrument becomes re-usable between deployments and even facilities.

instrument

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-05-22. Last modified 2026-06-25

ATTRIBUTES

 manufacturer : Private
Alias: 5-09 manufacturer

[Is static True. Containment is Not Specified.]

 model : Private
Alias: 5-09 model

[Is static True. Containment is Not Specified.]

 verticalRange : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) instrument
Cardinality: [0..1]

Target: Public (Class) ObservingMethod
Cardinality: [0..*]

 Association (direction: Unspecified)

Source: Public groups (Class) instrument
Cardinality: [0..1]

Target: Public belongs_to (Class) instrumentSet
Cardinality: [0..1]

 Association (direction: Unspecified) 5-09 instrument

Source: Public (Class) Deployment

Target: Public (Class) instrument

ASSOCIATIONS

Cardinality: [1..*]

Cardinality: [0..1]

instrumentSet

Class in package 'WMDR2-core'

instrumentSet

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-22. Last modified 2026-06-25

ATTRIBUTES

description : Private

[Is static True. Containment is Not Specified.]

id : Private

[Is static True. Containment is Not Specified.]

name : Private
Alias: title

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public groups (Class) instrument
Cardinality: [0..1]Target: Public belongs_to (Class) instrumentSet
Cardinality: [0..1]

ObservationSeries

Class in package 'WMDR2-core'

WMDR2 core JSON class: observation.

ObservationSeries

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES

applicationArea : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)Value(s) from <http://codes.wmo.int/wmdr/ApplicationArea>

[Is static True. Containment is Not Specified.]

contact : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES	
Observation-level contacts.	[Is static True. Containment is Not Specified.]
 id : Private Observation identifier, e.g. observation:<observedVariable>.	[Is static True. Containment is Not Specified.]
 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation keywords.	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation links.	[Is static True. Containment is Not Specified.]
 observedGeometry : Private cf. http://codes.wmo.int/wmdr/Geometry	[Is static True. Containment is Not Specified.]
 observedProperty : Private Alias: OMS:ObservedProperty, iop:Variable The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2. NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific. Examples include: - observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...) - objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll - observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text. - matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ... These elements will be supported by code lists from codes.wmo.int/wmdr2. Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc.	[Is static True. Containment is Not Specified.]
 officialStatus : str Private = primary Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Values from http://codes.wmo.int/wmdr/ObservationStatus	

ATTRIBUTES

Defaults to "primary" if not provided.

[Is static True. Containment is Not Specified.]

 **programAffiliation** : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Program-affiliation timeline fragments inherited from WMDR1 observations.

[Is static True. Containment is Not Specified.]

 **representativeness** : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

<http://codes.wmo.int/wmdr/Representativeness>

[Is static True. Containment is Not Specified.]

 **title** : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Human-readable title derived from domain, geometry, observed variable, and optional label.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1]

Target: Public has (Class) OfficialStatus
Cardinality: [0..1]

 Association (direction: Unspecified)

Source: Public (Class) ObservationSeries
Cardinality: [1]

Target: Public (Class) ObservedFeature
Cardinality: [1]

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1]

Target: Public reports_with (Class)
ReportingProcedure
Cardinality: [0..*]

 Association (direction: Unspecified)

Source: Public (Class) ObservationSeries
Cardinality: [1]

Target: Public (Class) ObservingMethod
Cardinality: [0..*]

 Association (direction: Unspecified)

Source: Public (Class) ObservationSeries

Target: Public (Class) ReferenceSurface

 Association (direction: Unspecified)

ASSOCIATIONS	
Source: Public groups (Class) ObservationSeries Cardinality: [1..*]	Target: Public belongs_to (Class) ObservationSeriesSet Cardinality: [0..*]
 Association (direction: Unspecified) 3-07 temporalGeometry	
Source: Public (Class) ObservationSeries Cardinality: [1]	Target: Public (Class) temporalGeometry Cardinality: [0..1]
describes	has_geolocatino
 Association (direction: Unspecified)	
Source: Public is_used_by (Class) ObservationSeries Cardinality: [1..*]	Target: Public uses (Class) Deployment Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservationSeries Cardinality: [1]	Target: Public observes_with (Class) ObservingProcedure Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservationSeries Cardinality: [1..*]	Target: Public uses (Class) Extension Cardinality: [0..1] ObservingProcedure(s)
 Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) Facility Cardinality: [1]	Target: Public hosts (Class) ObservationSeries Cardinality: [1..*]

ObservationSeriesSet

Class in package 'WMDR2-core'

grouping of observations (which are themselves time series)

ObservationSeriesSet
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Deployment temporal extent, date-resolution interval or null.
[Is static True. Containment is Not Specified.]

ATTRIBUTES	
 id : Private Deployment identifier; source XML id is preserved when available. [Is static True. Containment is Not Specified.]	
 name : Private [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public groups (Class) ObservationSeries Cardinality: [1..*] Target: Public belongs_to (Class) ObservationSeriesSet Cardinality: [0..*]	

ObservedFeature

Class in package 'WMDR2-core'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-feature-of-interest, of an observation.

ObservedFeature
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-10. Last modified 2026-06-25

ATTRIBUTES	
 domain : Private The largest scale in WIGOS, cf. http://codes.wmo.int/wmdr/Domain [Is static True. Containment is Not Specified.]	
 domainFeature : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Describes the type of the larger-scale environmental target or ultimate-feature-of-interest, of an observation. Concept motivated by requirements from hydrology, but may be useful elsewhere. New code list is proposed, with terms like - river basin - lake - river - catchment - canopy - epilimnion - hypolimnion - troposphere - tropopause - stratosphere	

ATTRIBUTES

- thermocline (useful as a general term, but needs "domain")
- boundary layer (useful as a general term, but needs "domain")
- ...

[Is static True. Containment is Not Specified.]

featureName : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

free text field, e.g., "Danube Basin", "Amazon rain forest", "Amazon river Basin"

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) ObservationSeries
 Cardinality: [1]

Target: Public (Class) ObservedFeature
 Cardinality: [1]

ObservingMethod

Class in package 'WMDR2-core'

ObservingMethod

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-26. Last modified 2026-06-26

ATTRIBUTES

 observingMethod : Private

from (new) unified code list of observing methods, based on
<http://codes.wmo.int/wmdr/ObservingMethodAtmosphere>
<http://codes.wmo.int/wmdr/ObservingMethodTerrestrial>
<http://codes.wmo.int/wmdr/ObservingMethodEarth>

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) ObservationSeries
 Cardinality: [1]

Target: Public (Class) ObservingMethod
 Cardinality: [0..*]

 Association (direction: Unspecified)

Source: Public (Class) instrument
 Cardinality: [0..1]

Target: Public (Class) ObservingMethod
 Cardinality: [0..*]

ObservingProcedure

Class in package 'WMDR2-core'

When are observations made? Describes an observingStrategy, and depending on it, a temporal schedule/coverage, e.g. "from Feb-Oct on working days from 06-18 UTC".
Implement as array of objects

ObservingProcedure
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-25

ATTRIBUTES	
 date : Private	
Date from when on the schedule is valid	[Is static True. Containment is Not Specified.]
 strategy : Private	
Values from http://codes.wmo.int/wmdr/SamplingStrategy	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservingProcedure Cardinality: [1]	Target: Public uses (Class) observingSchedule Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservationSeries Cardinality: [1]	Target: Public observes_with (Class) ObservingProcedure Cardinality: [0..*]

observingSchedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

May be expected for strategy "continuous" or "discrete"

observingSchedule
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-25

ATTRIBUTES
 @type : Private

ATTRIBUTES	
JSCalendar object type; const Event. [Is static True. Containment is Not Specified.]	
 duration : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Occurrence duration describing coverage window or sampling duration. [Is static True. Containment is Not Specified.]	
 recurrenceOverrides : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence override patches; null override values are invalid. [Is static True. Containment is Not Specified.]	
 recurrenceRules : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence rules. [Is static True. Containment is Not Specified.]	
 start : Private Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start. [Is static True. Containment is Not Specified.]	
 timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) IANA time zone or UTC default. [Is static True. Containment is Not Specified.]	
 uid : Private JSCalendar-safe schedule uid. [Is static True. Containment is Not Specified.]	
 wmi.int:samplingFrequency : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) The frequency at which raw / level 0 observations are obtained. Implement as {"value": int, "uom": str} [Is static True. Containment is Not Specified.]	
 wmo.int:aggregationInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Describes the smallest interval over which raw/ level 0 observations are aggregated and stored. Implement as {"value": int, "uom": str} [Is static True. Containment is Not Specified.]	

ATTRIBUTES

ASSOCIATIONS

 Association (direction: Unspecified)Source: Public describes (Class) ObservingProcedure
Cardinality: [1]Target: Public uses (Class) observingSchedule
Cardinality: [1..*]

OfficialStatus

Class in package 'WMDR2-core'

OfficialStatus

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-25. Last modified 2026-06-25

ATTRIBUTES

 date : Private

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)Source: Public describes (Class) ObservationSeries
Cardinality: [1]Target: Public has (Class) OfficialStatus
Cardinality: [0..1]

ProgramAffiliation

Class in package 'WMDR2-core'

ProgramAffiliation

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-25

Alias temporalProgramAffiliation

ATTRIBUTES

 date : Private

[Is static True. Containment is Not Specified.]

 programAffiliation : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

 programSpecificFacilityId : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES	
	[Is static True. Containment is Not Specified.]
 reportingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/ReportingStatus	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) ProgramAffiliation Cardinality: [0..1]

ReferenceSurface

Class in package 'WMDR2-core'

ReferenceSurface
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-26. Last modified 2026-06-26

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) ObservationSeries	Target: Public (Class) ReferenceSurface

reporting

Class in package 'WMDR2-core'

Reusable reporting template.

reporting
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-23

ATTRIBUTES	
 contact : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 dataFormat : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/DataFormat .	[Is static True. Containment is Not Specified.]

ATTRIBUTES

dataPolicy : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Value from <http://codes.wmo.int/wmdr/DataPolicy>.

[Is static True. Containment is Not Specified.]

internationalExchange : Boolean Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

levelOfData : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Value from <http://codes.wmo.int/wmdr/LevelOfData>.

[Is static True. Containment is Not Specified.]

numberOfObservationsInReportingInterval : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

referenceDatum : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Value from <http://codes.wmo.int/wmdr/CoordinateReferenceSystem>.

[Is static True. Containment is Not Specified.]

referenceTimeSource : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Single value or array, values from <http://codes.wmo.int/wmdr/ReferenceTime>.

[Is static True. Containment is Not Specified.]

timeliness : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Expected timeliness of reporting, an object with value and uom (default: seconds)

[Is static True. Containment is Not Specified.]

timeStampMeaning : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Aligned array, values from <http://codes.wmo.int/wmdr/TimeStampMeaning>.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 7-03 Temporal reporting

Source: Public describes (Class) ReportingProcedure

Cardinality: [1]

Target: Public uses (Class) reporting

Cardinality: [0..1]

ASSOCIATIONS

ReportingProcedure

Class in package 'WMDR2-core'

ReportingProcedure
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-23. Last modified 2026-06-25

ATTRIBUTES



date : Private

[Is static True. Containment is Not Specified.]



links : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

URL to related WIS record, entry point for data discovery and retrieval.

[Is static True. Containment is Not Specified.]



strategy : Private

[Is static True. Containment is Not Specified.]



uom : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Values from <http://codes.wmo.int/wmdr/unit>.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS



Association (direction: Unspecified)

Source: Public describes (Class) ReportingProcedure
Cardinality: [1]

Target: Public uses (Class) reportingSchedule
Cardinality: [1..*]



Association (direction: Unspecified) 7-03 Temporal reporting

Source: Public describes (Class) ReportingProcedure
Cardinality: [1]

Target: Public uses (Class) reporting
Cardinality: [0..1]



Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1]

Target: Public reports_with (Class)
ReportingProcedure
Cardinality: [0..*]

reportingSchedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

reportingSchedule

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-25. Last modified 2026-06-25

ATTRIBUTES	
◆ @type : Private JSCalendar object type; const Event.	[Is static True. Containment is Not Specified.]
◆ duration : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Occurrence duration describing coverage window or sampling duration.	[Is static True. Containment is Not Specified.]
◆ recurrenceOverrides : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence override patches; null override values are invalid.	[Is static True. Containment is Not Specified.]
◆ recurrenceRules : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence rules. To be used for "reportingInterval"; use start, duration and other elements to evaluate constraints if needed.	[Is static True. Containment is Not Specified.]
◆ start : Private Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start.	[Is static True. Containment is Not Specified.]
◆ timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) IANA time zone or UTC default.	[Is static True. Containment is Not Specified.]
◆ uid : Private JSCalendar-safe schedule uid.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) ReportingProcedure
Cardinality: [1]

Target: Public uses (Class) reportingSchedule
Cardinality: [1..*]

temporalGeometry

Class in package 'WMDR2-core'

temporalGeometry

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-25

Alias temporalGeometry

ATTRIBUTES

 coordinates : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Coordinate history array; each entry is lon, lat[, elevation].

[Is static True. Containment is Not Specified.]

 dates : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Date array aligned with coordinates.

[Is static True. Containment is Not Specified.]

 methods : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

use <http://codes.wmo.int/wmdr/GeopositioningMethod>

[Is static True. Containment is Not Specified.]

 type : str Private = MovingPoint

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) Facility
Cardinality: [1]

Target: Public has_geolocation (Class)
temporalGeometry
Cardinality: [0..1]

temporalGeometry

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalGeometry.

temporalGeometry
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-12. Last modified 2026-06-02

ATTRIBUTES	
coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation]. [Is static True. Containment is Not Specified.]	
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates. [Is static True. Containment is Not Specified.]	
methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) 3-07 temporalGeometry Source: Public (Class) ObservationSeries Cardinality: [1] describes	Target: Public (Class) temporalGeometry Cardinality: [0..1] has_geolocatino

Territory

Class in package 'WMDR2-core'

WMDS reference: 3-02

mandatory for fixed facilities, optional for mobile facilities.

Implemented as [{"date": date, "territory": str}, {}] like other historical<Name> classes.

Territory
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-23. Last modified 2026-06-25

ATTRIBUTES	
date : Private [Is static True. Containment is Not Specified.]	
territory : Private	

ATTRIBUTESAlias: **temporal Territory**Values from <http://codes.wmo.int/wmdr/TerritoryName>

[Is static True. Containment is Not Specified.]

ASSOCIATIONS Association (direction: Unspecified)Source: **Public (Class) Facility**
Cardinality: [1]Target: **Public (Class) Territory**
Cardinality: [1..*]